Factor Analysis Framework for a Systematic Long/Short Equity Portfolio

Pre-interview case response: Middle Office / Risk Management

Prepared by Lisheng | Presto Labs, Singapore | October 2026

Executive summary

I would build the framework around one principle: **risk management can see what a black-box model does, even when it cannot see how the model decides.** Positions, trades and P&L are observable at every timestamp. The framework therefore decomposes those, rather than trying to explain the prediction model itself.

The proposed framework has five components:

1. **A linear, holdings-based factor model as the core.** It uses factors that drive returns over hours to days (market, industry, short-term reversal, short-term momentum, liquidity, volatility, crowding, events), together with market-wide shocks such as volatility, liquidity, currency and rates moves, instead of the slow fundamental factors that dominate traditional models. A small number of statistical factors catch exposures nobody thought to define.
2. **A full P&L decomposition, not only a holdings attribution.** P&L is split into factor P&L, residual P&L, intraday trading P&L and execution costs. For strategies that hold positions for hours, the last two are often decisive.
3. **Machine learning as a diagnostic layer around the linear core.** Surrogate models, clustering and regime detection look for nonlinear or hidden exposures. Anything important they find is turned into an explicit factor in the linear model.
4. **Local models per market and asset class, linked by a global layer.** Each market (and crypto, if in scope) gets its own estimation, while shared factor definitions and a cross-model covariance layer allow firm-level aggregation.
5. **Validation on two tests.** The statistical test asks whether risk forecasts are calibrated and attribution reconciles to P&L. The practical test asks whether the framework changed a decision or flagged a problem before it cost money.

I would build the model fully in-house. The core factors come almost entirely from price, volume and position data the firm already collects for trading, so no vendor licence is needed, and a simple statistical (PCA) model serves as an independent challenger. The framework is rolled out in phases so that risk managers and PMs get value within weeks rather than after a long build.

1. How can a factor framework help the firm?

A factor framework gives the firm a **common language for risk**. Without one, each team describes its book in terms of its own signals, which cannot be compared or added up. With one, every strategy, whatever model generated it, can be expressed as exposures to the same set of factors plus a residual. That shared language is valuable to all three stakeholder groups, but each needs different answers from it.

1.1 Risk Management / Middle Office

For risk management, the framework turns a set of opaque strategies into measurable, limitable exposures. The questions it should answer:

- **What is each strategy actually exposed to, and has that changed?** Black-box models can drift after a retrain or a regime change without anyone deciding it. Daily exposure monitoring catches unintended bets, such as a book that is dollar-neutral but beta-long, or one that has quietly turned into a short-term reversal strategy.
- **Where are the hidden concentrations across teams?** Two teams with uncorrelated historical P&L can still hold the same factor tilt. Aggregation shows whether teams offset each other or stack the same risk.
- **What should the limits be, and on what?** Limits on beta-adjusted net exposure, industry and style factor exposures, predicted volatility and concentration are more meaningful than gross and net dollar limits alone.
- **How would the book behave under stress?** Historical and hypothetical scenarios (factor crashes, liquidity shocks, deleveraging) need factor exposures as inputs.
- **Is a drawdown explained by factors or by the strategy itself?** The answer decides whether the response is hedging, de-risking or a model review.

Decisions supported: setting and enforcing limits, escalation, risk reduction during stress, and approval of new strategies (does a new strategy add a new source of return or repeat an existing risk?).

1.2 Quant Traders / Portfolio Managers

PMs will only use the framework if it helps them make money, so it should be designed to answer their questions rather than as a compliance tool.

- **Is my P&L coming from the alpha I intended, or from factor exposures I could get cheaply?** A strategy whose returns are mostly explained by short-term reversal is effectively a reversal strategy, however sophisticated the model.
- **Which unintended exposures should I neutralise?** Unwanted factor exposures can often be hedged or constrained in the optimizer at low cost to alpha.
- **Did the new model version change my risk profile?** Comparing exposures before and after a retrain is a cheap, effective check on black-box models.
- **Why am I in a drawdown?** Separating factor headwinds from alpha decay prevents two opposite mistakes: shutting down a working model during a bad factor regime, and blaming factors while the alpha has actually disappeared.
- **How much of my gross alpha survives costs?** For high-turnover strategies, execution cost is often the largest item the PM can control.

Decisions supported: optimizer constraints, hedging, model release approval, capacity sizing and research priorities.

1.3 Senior Management / Capital Allocation

- **How diversified are we really?** P&L correlations combined with overlapping factor exposures give an effective number of independent bets. Ten teams can turn out to be three bets.

- **Are we paying for alpha or for factor exposure?** Residual (factor-adjusted) Sharpe ratio is a better measure of skill than raw Sharpe ratio. Factor returns can be obtained more cheaply and behave differently under stress.
- **Where should the next unit of capital go?** Allocation should consider residual Sharpe, correlation with the existing book, capacity and cost sensitivity, not raw historical P&L alone.
- **What is the firm-level tail risk?** Firm-wide exposure to a crowded factor or a market-wide liquidity shock matters more than any single team staying within its own limits.

Decisions supported: capital and risk budgets per team, hiring and research investment, strategy shutdowns, and the firm's overall risk appetite.

Summary by stakeholder

Stakeholder	Core question	Decisions supported	Key outputs
Risk / Middle Office	What are we exposed to, where is it concentrated, and what happens under stress?	Limits, escalation, de-risking, new strategy approval	Exposure monitor, limit usage, stress results, drift alerts
Quant Traders / PMs	Is my P&L the alpha I intended, net of costs?	Hedges, optimizer constraints, model releases, capacity	Daily P&L attribution, release comparison, cost breakdown
Senior Management	How many independent bets do we have, and where should capital go?	Capital and risk budgets, shutdowns, risk appetite	Residual Sharpe by team, overlap map, firm-level exposures

Trade-offs

One framework, different users. Risk wants stability and comparability, PMs want precision for their own strategy, and senior management wants simplicity. Separate models for each group would produce numbers that disagree and erode trust. I would use **one common core model as the single source of truth**, with different views and drill-downs for each audience.

Simplicity versus completeness. Too many factors make attribution unstable and hard to explain; too few leave a large unexplained residual. The factor set should stay small and only grow when a new factor proves its value (see Section 2).

> **How I would judge usefulness for this part**
>
> The framework is useful if specific decisions changed because of it: a PM added a constraint, capital moved between teams, a limit breach or crowded overlap was caught early, or a drawdown was explained within a day.
>
> Usage is a secondary signal: do PMs and management open the reports without being asked, and do they request new views?

2. Which factors are relevant?

2.1 Selection principles

A risk model needs factors that drive **co-movement over the holding horizon**. Many traditional factors carry long-run return premia but contribute little to day-to-day variance. Given holding periods of hours to

days, high turnover, diversified universes, low net exposure and ML-based models, a factor earns a place in the core model if it meets four tests:

- It explains a meaningful share of short-horizon (daily or intraday) return covariance, not only long-horizon expected returns.
- Its exposures can be computed and refreshed at the trading frequency.
- It is interpretable and actionable: someone can hedge it, limit it or decide to keep it.
- It is not redundant with factors already in the model.

2.2 Proposed core factor set

Most factors are **cross-sectional**: stock characteristics whose returns are estimated by regression each period. Factors marked **(time-series)** are market-wide shocks, such as a volatility spike or a currency move; a stock's or strategy's exposure to them is estimated from its sensitivity to the shock.

Factor	Typical definition	Why it matters for these strategies
Market / beta	Market return (index futures returns intraday); stock exposure = predicted beta	Low dollar net is not low beta net. A book long low-beta and short high-beta stocks is net short the market. Intraday, futures lead the cash market and are the hedging instrument.
Country and industry	Membership indicators from an available classification (exchange or data-provider sectors), or return-correlation clusters	The largest source of short-horizon co-movement. Country factors capture market-level moves in multi-market books.
Short-term reversal	Return over the past 1 to 5 days, or intraday	The dominant short-horizon effect in equities. ML models trained on recent prices often load heavily on it without anyone specifying it, and many quant firms share it.
Short-term momentum, overnight vs intraday	Recent intraday trend; overnight gap vs intraday return	Overnight and intraday returns behave differently and are traded by different participants, so they should be separate factors.
Liquidity and turnover	ADV, turnover, spread, Amihud illiquidity, relative tick size (tick divided by price)	High-turnover strategies tilt toward liquid names, and liquidity shocks hit all short-horizon strategies at once. Also drives cost and capacity.
Volatility	Recent realized or residual volatility	Captures risk-on/risk-off rotation. ML models often favour volatile names because they offer larger predicted moves.
Size	Log market capitalisation	Partly a liquidity proxy; kept because long/short books can carry unintended size tilts.
Short-side crowding	Composite of short interest as a share of float, days-to-cover, and prime broker borrow fee and availability	Captures squeeze risk, when forced covering hits everyone short the same names at once (e.g. January 2021). The only crowding measure in the core regression; see below.

Factor	Typical definition	Why it matters for these strategies
Events and calendar	Earnings windows, index rebalances, option expiry, corporate actions, macro announcements, and an "in-play" flag for unscheduled news (abnormal volume plus a price jump)	Short-horizon returns and risk cluster around events; a strategy may be an event strategy without being labelled as one. The in-play flag covers news risk without a news feed.
Volatility and liquidity shocks (time-series)	Changes in implied volatility indices (VIX or local equivalents) and in aggregate spreads or order book depth	Reversal strategies are effectively paid to provide liquidity and lose when market liquidity disappears (August 2007, March 2020). The cross-sectional liquidity factor does not capture this.
Currency, rates and prior-session return (time-series)	Stock betas to local currency, rates futures and, for Asian and European markets, the previous US session return	Exporters and domestic stocks move apart with the currency within a day (e.g. Japan in August 2024); rates surprises move sectors within minutes; Asian opens partly reflect the US session.
Statistical residual factors	A few principal components of the named model's residual returns	Catch common exposures nobody defined, which matters most when the alpha models are black boxes.

Crowding at three levels

Crowding is not a single factor. I would measure it at three levels, each built differently, and only the first enters the core regression.

1. Short-side crowding (core factor). The inputs for each stock are short interest as a share of free float, days-to-cover (short interest divided by average daily volume), prime broker borrow fee and availability, and securities lending utilisation where available. Borrow data comes from Presto's own prime brokers daily as part of shorting, so it is the freshest crowding data available and needs no extra licence. Construction:

1. **Lag every input to its publication date**, not the date it refers to, to avoid look-ahead bias.
2. **Transform skewed inputs** with logs or ranks, then winsorise the extremes.
3. **Standardise within each market** to a mean of zero and unit spread.
4. **Average the available inputs into one composite.** A stock with no data scores zero (neutral) rather than being dropped.
5. **Remove the overlap with size, volatility and liquidity**, because crowded shorts tend to be small, volatile and illiquid. Regress the composite on those exposures each day and keep the residual, in a fixed, documented order.
6. **Include it in the daily cross-sectional regression** like any other style factor. Its return is effectively heavily shorted stocks against the rest, after controlling for everything else.

The factor earns its place only if its return spikes in known squeeze episodes, it carries material volatility, and adding it reduces residual correlation across teams. Otherwise it moves to the monitoring layer.

2. Factor-level crowding (monitoring indicator). Whether too many investors run the same strategy, such as short-term reversal, is a property of a factor over time rather than a stock characteristic, so it does not belong in a cross-sectional regression. Following Lou and Polk's comomentum measure, I would take the stocks in the top and bottom deciles of each core factor and track the average pairwise correlation of their residual returns over roughly the past 60 days. Rising correlation within a factor's long or short leg suggests

capital is crowding into that trade. Short interest concentrated in a factor's short leg is a complementary signal (Hanson and Sunderam). These indicators drive alerts and scale up stress tests for factors that look crowded.

3. Internal crowding (liquidity stress). With several teams trading similar universes, the firm can crowd itself. This is measured entirely from internal data: the firm-wide position in each stock as a share of its average daily volume (days to unwind), the number of teams holding the same name on the same side, and the overlap between teams' position vectors. These feed liquidity stress tests that estimate the cost if every team had to exit at once. No single team can see this risk, which makes it one of the clearest contributions of an independent middle office.

Downgraded measures. US 13F hedge fund ownership is quarterly, lagged and long-only, so it is only a background monitor. ETF ownership and flows are monitored around index and ETF rebalances but are not a core factor. Short-interest data differs by market in frequency, lag and level of disclosure; where short selling is restricted, the short-side factor is not meaningful and the framework relies on internal and return-based measures instead.

Additional factors for intraday strategies

For strategies holding for minutes to hours, a purely cross-sectional model misses market-wide shocks. The framework therefore combines cross-sectional factors with time-series factors (Section 3.1), and adds the following.

High priority: core intraday model.

- **Volatility and liquidity shocks.** Nagel (2012) shows that short-term reversal returns behave like payment for providing liquidity: they are higher when implied volatility is high and suffer when liquidity evaporates. A book of reversal strategies can be short volatility and short liquidity while every cross-sectional exposure looks neutral. Factors are built from changes in an implied volatility index (or index futures realized volatility where none exists) and in aggregate spreads or depth; each strategy's exposure comes from rolling regressions of its P&L on these shocks. Used for stress testing and limits.
- **Intraday market factor from index futures.** Futures lead the cash market within the day and are how intraday beta is hedged, so futures returns define the intraday market factor. Betas must be estimated with care because of the Epps effect (Section 3.4).
- **Currency, rates and prior-session factors.** Rolling time-series betas of each stock to the local currency, rates futures and the previous US session return, shrunk toward the industry average because individual estimates are noisy. Macro announcements (inflation data, central bank decisions) are added to the events calendar.

Medium priority: test for the core, otherwise monitor.

- **ETF and index-basket co-movement.** ETF and index-futures arbitrage trades whole baskets at once, and stocks with more ETF ownership are more volatile (Ben-David, Franzoni and Moussawi, 2018). Exposure is ETF ownership as a share of shares outstanding, from public ETF holdings, plus index-membership flags; Nikkei 225 membership in Japan is a clear example. This is a different use of ETF data from the crowding monitor.
- **Order flow imbalance.** Buy-initiated minus sell-initiated volume explains much of short-term price movement (Chordia and Subrahmanyam, 2004; Cont, Kukanov and Stoikov, 2014). Because it is close to an alpha signal and to execution analysis, it belongs in the intraday attribution model only, not the daily risk model.

- **Retail flow and attention.** Important in retail-heavy markets (China A-shares, Korea, Taiwan) and in US meme episodes. US retail trades can be identified from sub-penny price improvement (Boehmer, Jones, Zhang and Zhang, 2021); elsewhere, abnormal volume and the share of small trades are proxies. Linked to the crowding layer.
- **Relative tick size.** Stocks whose tick is large relative to their price have spreads pinned at one tick and different intraday cost and reversal behaviour. Added as a component of the liquidity factor rather than a standalone factor.

Deliberately excluded. Intraday seasonality patterns, such as returns repeating at the same half-hour across days (Heston, Korajczyk and Sadka, 2010) or the market's first half-hour predicting its last (Gao, Han, Li and Zhou, 2018), are return predictability, which makes them alpha rather than risk; they are handled by normalising for seasonality. Order book features at sub-second horizons are too fast for a risk model and belong to execution analysis.

Every candidate must still pass the selection tests in Section 2.1 on Presto's own data. I would expect the volatility and liquidity shock factors, the futures-based market factor and the currency betas to pass most clearly.

2.3 Traditional factors I would treat as less relevant

Value (book-to-price, earnings yield), quality and profitability, investment and asset growth, dividend yield, leverage, growth, and long-horizon (12-1 month) momentum are designed for investors who hold positions for months. Their characteristics change quarterly, their premia accrue slowly, and they explain little of a short-horizon strategy's daily P&L.

Less relevant does not mean irrelevant. I would keep these as **monitoring and stress factors** rather than core attribution factors, for three reasons:

- Long/short books can carry unintended tilts to them; ML models may systematically favour low-quality, volatile names, for example.
- In factor rotations and deleveraging events, slow factors can move violently in a single day, as in the momentum/value reversal on 9 November 2020. Short-horizon books are affected through crowding and liquidity.
- Senior management and investors will ask about them.

Trade-off: including them in the core model adds collinearity and noise to daily attribution; excluding them entirely creates blind spots in stress testing. A separate monitoring layer is the compromise.

How the monitoring layer works

The monitoring layer is **not a second risk model**. It has no covariance matrix, specific risk model or limits of its own. It adds two light components on top of the core model, which remains the only source of official attribution, risk forecasts and limits.

Exposures. Each stock is scored on the slow characteristics, standardised the same way as the core factors. A strategy's exposure is its holdings multiplied by those scores, which requires no regression. Price-based slow factors (12-1 month momentum, long-term reversal) can start immediately; fundamental ones (value, quality) are added once a point-in-time fundamentals feed is available.

Factor returns from a second-stage regression. The core regression runs as normal; its residuals are then regressed on the slow-factor exposures, after first removing from each slow exposure the part explained by the core exposures:

$$\text{Stage 1:} \quad r = X_{core}\, f_{core} + u$$
$$\text{Stage 2:} \quad u = X_{slow}\, g + e$$
$$P\&L = \text{core factor } P\&L + \text{slow-factor } P\&L + \text{remaining specific } P\&L$$

Core attribution is untouched, slow factors only receive credit for what the core cannot explain, and the decomposition still adds up. Simple top-minus-bottom characteristic portfolios are a quick alternative, but they overlap with the core factors and are not additive.

The layer is used in four ways:

- **Exposure alerts** with soft thresholds on the level of, and sudden changes in, slow-factor exposures (for example after a model retrain).
- **Stress testing**, applying historical slow-factor moves such as 9 November 2020 to current exposures. This complements the most robust stress test, which needs no model: applying the actual stock-level returns of a past stress day to today's holdings.
- **Crowding indicators.** Return-based crowding measures for each core factor (described in Section 2.2) sit in the same layer and drive alerts and stress scaling.
- **Promotion.** If a slow factor consistently explains a meaningful share of strategy residuals or stress losses, it is promoted into the core model through the governance process.

2.4 Should factors differ by market, horizon or strategy?

By market: yes, with a common taxonomy and local estimation. Factor definitions should be the same everywhere so the firm can aggregate, but each market should be estimated separately because market structure differs: trading hours, tick sizes, retail participation, short-selling constraints, price limits and settlement rules (for example, daily price limits and the restriction on selling same-day purchases in China A-shares), and local sector structures. Some markets need extra factors, such as limit-hit indicators in China or foreign flows in some Asian markets.

By horizon: two reconciled layers. A daily model serves risk forecasting, limits and aggregation. An intraday model (bar-level returns, index futures as the market factor, intraday seasonality, auctions, overnight versus intraday split, and time-series shock factors) serves attribution for strategies that hold for hours. Intraday attribution summed over a day should reconcile to the daily view.

By strategy: one common core, optional supplements. Every strategy is measured against the same core model, because that is what makes aggregation and comparison possible. Strategy-specific supplementary factors (event factors for an event-driven team, for example) can appear in drill-down views, but never replace the core.

Trade-off: a bespoke model per team would fit each strategy better, but the firm would lose the ability to add risks up across teams. Consistency matters more for a risk function than marginal fit.

2.5 Crypto, if in scope

Presto also trades crypto, so the framework should anticipate it. I would build a **separate local crypto model** because:

- The factor structure is different: BTC and ETH beta explain a much larger share of variance than the market factor does in equities.
- Crypto has its own factors: funding rates, perpetual-spot basis, exchange-specific liquidity and venue risk, token supply events such as unlocks, stablecoin flows, and size or "alt-season" effects.
- It trades 24/7, with a different volatility scale and fatter tails. A shared covariance estimate would be dominated by crypto.

The crypto model links to the equity models through the global layer described in Section 3.2. Equities with crypto exposure (exchanges, miners, companies holding crypto on their balance sheet) need explicit loadings on the crypto factors. Stress tests should include cross-asset contagion episodes such as LUNA (May 2022), FTX (November 2022) and the August 2024 carry-trade unwind.

> **How I would judge usefulness for this part**
>
> Factor relevance is an empirical question, so each factor should prove its place: marginal contribution to explained cross-sectional variance, significance and stability of its returns, whether adding it reduces residual correlation across teams, and whether PMs can act on it.
>
> Factors that fail these tests are moved to the monitoring layer or dropped through a documented governance process.

3. How would I implement the framework?

3.1 Architecture

Layer	What it does	Key output
1. Data	Positions, trades, market, reference and execution data	Clean, time-stamped panel tagged by strategy and team
2. Exposures	Standardised factor exposures for every stock	Exposure matrix X
3. Factor returns	Cross-sectional regression each period	Factor returns f and residual returns u
4. Risk model	Factor covariance and specific risk	Volatility forecasts, marginal risk contributions
5. Attribution	Factor, residual, trading and cost decomposition	P&L attribution by strategy, team and firm
6. Diagnostics	Statistical factors, ML surrogates, clustering	Hidden exposures and overlap between teams
7. Reporting	Dashboards, reports, alerts	Views tailored to each stakeholder

The core model is the standard linear factor structure. For stock returns r, exposures X, factor returns f and specific returns u:

$$r = X f + u$$
$$Var(R_p) = w^T X F X^T w + w^T \Delta w$$
$$Factor\ P\&L = (X^T w)^T f$$

Here w is the vector of holdings, F the factor covariance matrix and Δ the diagonal matrix of specific variances. The portfolio's factor exposure is $X^T w$.

Time-series factors enter the same structure. Their exposures are not observed characteristics but estimated sensitivities B to observed shock series g, such as changes in implied volatility, currency returns or index futures returns:

$$r = Xf + Bg + u$$

Because g is observed directly, only B needs estimating, and the decomposition remains additive. The covariance matrix F is extended to include the shock series.

3.2 Main methodological choices

Choice	Options	My recommendation	Trade-off
Attribution basis	Holdings-based vs returns-based	Holdings-based as primary; returns-based regression of strategy P&L as a cross-check	Holdings-based needs a full position pipeline but is precise and timely. Returns-based needs only P&L but is noisy and lags exposure changes.
Model type	Characteristic (cross-sectional), time-series, statistical, hybrid	Hybrid: cross-sectional characteristic factors, time-series shock factors, and a few statistical residual factors	Named factors are interpretable but miss unknown exposures, and cross-sectional factors miss market-wide shocks. Time-series betas are noisy and need shrinkage; statistical factors are harder to explain and less stable.
Functional form	Linear vs nonlinear (ML)	Linear core, ML as a diagnostic layer	Linear gives additive, auditable attribution but misses nonlinear effects, which the diagnostic layer covers.
Build vs buy	Vendor model vs in-house	In-house, with a statistical (PCA) model as an independent challenger	In-house fits our horizons, gives full control and avoids licence costs, but takes build effort and has no external audit, so validation must carry more weight.
Frequency	Daily vs intraday	Daily risk model plus intraday attribution	Intraday adds noise, data volume and microstructure issues; daily alone misses P&L from strategies holding for hours.
Cross-market structure	Single global model vs local models with a link	Local models linked by a global covariance layer	A single model is simpler but fits each market poorly. The tiered model needs a reconciliation layer.

Why a linear core rather than deep learning

The black-box nature of the alpha models does not require a black-box risk model. I would keep the core linear for five reasons:

- **Additivity.** Exposures and P&L contributions sum cleanly across strategies, teams and the firm, which limits and aggregation depend on.
- **Interpretability.** "Team A is 0.4 standard deviations long short-term reversal" is a statement a PM, a CRO and a capital allocator can all act on.
- **Stability.** Deep-learning attributions are non-additive and can change materially across retrains, which makes trends and limits meaningless.

- **Low signal-to-noise.** Factor returns are noisy, so a flexible model tends to overfit the risk model itself.
- **Independence.** The risk model should be a different, simpler lens than the models it monitors, not a second black box auditing the first.

Nonlinearity can still be handled inside the linear framework through interaction terms (for example, reversal × liquidity), regime-conditional betas and volatility-scaled exposures.

Where ML adds value is discovery:

- A surrogate model, such as gradient boosting with SHAP values, mapping each strategy's position weights to stock characteristics. It reveals what the strategy is effectively selecting on, including conditional exposures such as reversal only in illiquid names.
- Clustering of strategy P&L and position vectors to find overlap between teams.
- Regime detection, to adjust risk forecasts when market conditions change.

The rule: anything the ML layer finds that matters is turned into an explicit factor or interaction term in the linear model, so it becomes measurable and limitable.

The rule also guards against known weaknesses of post-hoc explanations. Shapley-based attributions depend on how "missing" features are simulated and can credit features the model does not use, and perturbation-based explainers can be fooled by models that recognise their synthetic inputs (see Appendix B). An explicit factor, by contrast, is measured directly from holdings with a fixed definition.

Linking local models (cross-market and crypto)

Each market or asset class has its own local model. A global layer then estimates the covariance between local factor returns, similar in spirit to the integrated global models used in commercial risk systems. Firm-level risk is computed by combining local exposures with this cross-model covariance. The main estimation challenge is non-synchronous trading, which is addressed in Section 3.4.

3.3 Data requirements

Data	Used for	Frequency	Challenges
Positions and trades from OMS/EMS, tagged by strategy and team	Exposures, attribution, trading P&L	Real-time / per fill	Must reconcile to the official books; consistent strategy tagging
Market data: bars, closes, auctions, quotes	Returns; reversal, momentum, volatility and liquidity exposures	Intraday and daily	Corporate-action adjustment, bad ticks, trading halts
Reference data: listings, classifications, corporate actions, index membership	Industry and country factors, universe definition	Daily	Survivorship-free history; licensed schemes such as GICS may be unavailable, so use exchange or data-provider sectors, or correlation-based clusters
Fundamentals	Monitoring factors such as value and quality (size needs only price and shares outstanding)	Daily / quarterly	Point-in-time data to avoid look-ahead bias; start with price-based proxies if no feed exists

Data	Used for	Frequency	Challenges
Short interest, prime broker borrow fees and availability, lending utilisation; ownership and ETF data	Short-side crowding factor; ownership and ETF data for monitoring only	Daily (borrow) to quarterly (13F)	Coverage, lags and disclosure vary by market; borrow data comes from our own prime brokers
Company and macro event calendars	Event factors	Daily	Point-in-time earnings dates; macro announcement times to the minute
Index futures, currencies, rates futures, implied volatility indices	Intraday market factor; volatility, currency and rates shock factors	Intraday	Aligning timestamps across venues; local implied volatility indices do not exist in every market
Trades and quotes with trade direction	Order flow imbalance, retail flow, tick size	Intraday	Large data volume; trade-direction classification is approximate
ETF holdings and index membership	ETF and index-basket co-movement	Daily	Public but scattered across issuers; coverage varies by market
Execution data: fills, decision and arrival prices, fees, borrow, financing	Cost attribution	Per fill	Needs decision timestamps from the strategy to measure slippage properly
Crypto: spot and perpetual prices by venue, funding, basis, open interest	Crypto model	Intraday	Venue fragmentation; aligning 24/7 data with exchange hours

The position and execution pipeline is usually the hardest part to build and the most valuable. I would build it first, because a sophisticated model on an unreconciled position feed loses credibility quickly.

3.4 Estimation issues

- **Responsiveness versus noise.** Short-horizon risk changes quickly. I would use exponentially weighted estimates with short half-lives for volatilities and longer half-lives for correlations, plus a volatility regime adjustment that scales forecasts when recent realised factor volatility departs from the forecast.
- **Noisy covariance matrices.** Optimised portfolios systematically load on directions where the covariance matrix underestimates risk. Shrinkage (such as Ledoit-Wolf) and eigenfactor adjustments reduce this bias.
- **Collinearity.** Reversal and momentum, size and liquidity, and volatility and beta overlap. Exposures should be winsorised and standardised; any orthogonalisation should follow a fixed, documented order; and variance inflation factors should be monitored. Industry factor returns are constrained (for example, cap-weighted to sum to zero) so the market factor stays identified.
- **Regression weighting.** Weighted least squares, with weights proportional to the square root of market cap or to liquidity, limits the influence of noisy small caps. The estimation universe should match the traded universe.
- **Specific risk.** Short histories and fat tails make stock-level estimates unreliable. A structural model that shrinks toward peers with similar size, volatility and industry, plus extra risk on event days, is more robust.

- **Non-synchronous trading.** Markets close at different times and crypto trades continuously. Cross-market correlations need overlapping windows, lagged-correlation adjustments or longer aggregation periods.
- **Intraday effects.** Intraday volatility is U-shaped, auction prints behave differently and overnight gaps are large. Overnight and intraday returns should be modelled separately and normalised for intraday seasonality.
- **Asynchronous trading within the day (Epps effect).** Correlations and betas estimated from very high-frequency returns are biased toward zero, because stocks do not trade at the same instants, so a book can look beta-neutral while carrying real intraday beta. Remedies are sampling at 15 to 30 minutes, lagged-beta corrections, or estimators built for asynchronous data such as Hayashi-Yoshida.
- **Noisy time-series betas.** Betas to currency, rates and volatility shocks are estimated from short rolling windows and are unstable for individual stocks. Shrinking them toward the industry average and checking their stability over time keeps exposures usable.
- **Data hygiene.** Corporate actions, halts, stale prices, survivorship bias and look-ahead in fundamental and event data all need point-in-time handling.

3.5 P&L attribution design

For short-holding-period strategies, a holdings-only attribution based on daily snapshots misses much of the P&L. I would decompose total P&L as:

Total P&L = Factor P&L + Residual P&L + Intraday trading P&L
− Execution costs − Financing and borrow

- **Factor P&L:** start-of-period exposures multiplied by factor returns, summed over factors.
- **Residual P&L:** holdings multiplied by specific returns. This is the closest proxy for alpha, but it also contains model error.
- **Intraday trading P&L:** P&L from positions opened and closed between snapshots, computed from fills as the difference between actual P&L and the P&L of holding the start-of-period portfolio.
- **Execution costs:** slippage against the decision price, market impact, commissions and fees.

Attribution must **reconcile to the official P&L within a tolerance every day**, and the unexplained difference is itself a monitored metric. More frequent snapshots shrink the trading P&L bucket but increase noise and computation, so snapshot frequency should match each strategy's holding period. Multi-period attribution needs a consistent linking method so the components add up over weeks and months.

3.6 Validation approach

Statistical validity

- **Bias statistics.** The standard deviation of realised returns divided by forecast volatility should be close to 1, within roughly $1 \pm \sqrt{(2/T)}$ for T observations. I would test this on factors, stocks and, most importantly, the actual strategy portfolios, because optimised portfolios expose model biases that stock-level tests miss.
- **Out-of-sample explanatory power.** Cross-sectional R^2 of the factor regressions, and the share of each strategy's P&L variance explained by factors.
- **Factor quality.** Significance and stability of factor returns, and collinearity checks.

- **Exposure stability.** A strategy's exposures should not jump unless its positions did. Results should not be overly sensitive to estimation choices.

Missing-factor tests

- **Residual correlation across teams.** If residual returns are materially correlated across teams, there is a common factor the model is missing. This is the most useful single test for a multi-team firm.
- **PCA on residuals.** Dominant components are candidates for new factors.
- **Shared residual shocks.** Large residual losses hitting several teams on the same day usually indicate crowding or deleveraging that the model does not capture.

Historical event replay

Replay known stress episodes: the August 2007 quant deleveraging, March 2020, the November 2020 momentum/value reversal, the January 2021 short squeeze and the August 2024 carry-trade unwind, plus May and November 2022 for crypto. Two questions for each: would the exposures and forecasts have flagged the risk beforehand, and does the attribution explain the losses afterwards?

Benchmark comparison and governance

Without a vendor model, the in-house model needs challengers of its own. It should beat both a simple baseline (market plus industry only) and a purely statistical PCA model on calibration and explanatory power; otherwise its complexity has not earned its place. Public academic factor data, such as Kenneth French's data library, provides an external sanity check that in-house definitions of slow factors behave as expected, mainly for the US and other developed markets.

Ongoing governance includes a monthly validation report, versioning and a change log for factor definitions, independent review, and clear triggers for recalibration. Because there is no external audit of the model, independent review within the firm matters more.

3.7 Presenting outputs

Audience	Format and frequency	Core content
Risk / Middle Office	Intraday dashboard, daily risk report, alerts	Exposures against limits, predicted volatility, top risk contributors, limit breaches, stress results, exposure drift, internal crowding (firm-wide position against volume, team overlap)
Quant Traders / PMs	Daily attribution report, on-demand drill-down, model release comparison	Factor, residual, trading and cost P&L; exposure history; before/after comparison for model releases; hedging suggestions
Senior Management	Weekly one-page summary, monthly capital review	Firm-level exposures, residual Sharpe by team, overlap map, effective number of bets, risk budget usage, stress scenarios

Design principles for the outputs:

- **One source of truth.** Every view rolls up from the same numbers, so risk, PMs and management never argue about whose figures are right.
- **Drill-down.** Users can move from firm to team to strategy to stock.
- **Show uncertainty.** Residual Sharpe ratios carry confidence intervals, and short-window attribution is clearly labelled as noisy.

- **Stable definitions.** Factor definitions change rarely, with notice and documentation, so that trends remain meaningful.
- **Lead with the question.** Each report opens with the decision it supports, not with raw data.

3.8 Phased rollout

Phase 1: Foundations. Build the position, trade and P&L pipeline with strategy tagging; build an in-house baseline model (market, industry, country, size, beta, liquidity, short-term reversal) and a PCA challenger; report basic exposures; produce daily attribution that reconciles to official P&L. This delivers value to risk immediately.

Phase 2: Short-horizon fit. Add custom factors (reversal, intraday momentum, overnight/intraday split, crowding, events), the futures-based intraday market factor, volatility, liquidity, currency and rates shock factors, intraday attribution and execution-cost decomposition. This delivers value to PMs.

Phase 3: Discovery. Add statistical residual factors, the ML diagnostic layer, cross-team overlap analysis, tests of the medium-priority intraday factors (ETF and basket co-movement, order flow, retail flow), and the crypto model with the global linking layer.

Phase 4: Decision integration. Introduce factor-based limits, systematic stress testing, inputs to capital allocation, and optimizer integration agreed with PMs.

The ordering is deliberate: trusted, reconciled data in front of users comes before sophistication.

4. How to judge whether the framework is useful in practice

Statistical validity is necessary but not sufficient. A perfectly calibrated model that nobody uses has failed. I would track a two-part scorecard.

Dimension	Metric	Good signal
Statistical		
Calibration	Bias statistics on strategy portfolios	Within the confidence band most of the time
Explanatory power	Share of strategy P&L variance explained by factors	Stable over time; residuals not correlated across teams
Reconciliation	Attribution vs official P&L	Unexplained difference within tolerance every day
Stability	Exposure changes not caused by position changes	Rare and explainable
Practical		
Early warning	Concentration, crowding or drift flagged before losses	Documented cases
Decision impact	Limits, hedges, constraints or capital moves informed by the framework	Documented decisions
Adoption	Unprompted use of reports; requests for new views	Rising over time
Drawdown explanation	Large losses explained (factor, residual or cost) within a day	Most large losses

Dimension	Metric	Good signal
Cost-benefit	Effect of hedging or neutralising flagged exposures	Better risk-adjusted returns at acceptable alpha cost

> **The test I would apply after six months**
>
> Can we name specific decisions that were made differently because of the framework, and losses that were avoided or understood faster? If not, the framework needs to change, however good its statistics look.

5. Summary of key trade-offs

Decision	My choice	What we gain	What we give up
Linear vs deep learning	Linear core with ML diagnostics	Additive, interpretable, stable attribution	Some nonlinear effects, partly recovered through diagnostics and interaction terms
Named vs statistical factors	Hybrid	Interpretability plus coverage of unknown exposures	Statistical factors are harder to explain
Vendor vs in-house	In-house, with a PCA challenger	Horizon fit, full control, no licence cost	Build effort; no external audit, so validation carries more weight
Daily vs intraday	Both, reconciled	Complete picture for short holding periods	Complexity and data volume
Cross-sectional vs time-series factors	Both (hybrid)	Captures market-wide shocks that stock characteristics miss	Time-series betas are noisy and need shrinkage
One model vs per strategy	Common core with supplements	Aggregation and comparability	Some fit for individual strategies
Global vs local	Local models with a global link	Local fit and firm-level aggregation	Reconciliation effort; non-synchronous data issues
Slow traditional factors	Monitoring layer (exposures plus second-stage regression), not core	Cleaner daily attribution; stress coverage	Slow factors only get credit for what the core cannot explain

Appendix A: Likely follow-up questions

If the residual is large, does that mean the model is bad?

Not necessarily. For alpha strategies the residual should be large, because that is where the alpha sits. The warning signs are residuals that are correlated across teams, or residual losses that cluster in crowded or deleveraging episodes.

What if a team says the factor model does not describe their strategy?

Test it together using their returns and positions. If they are right, add supplementary factors in their drill-down view, and promote a factor to the core if it passes the selection tests. The common core stays, because aggregation depends on it. Their buy-in is itself one of the usefulness criteria.

Why not use returns-based style analysis? It is simpler.

It is noisy, lags changes in exposure and cannot see intraday trading. It is a good cross-check, and useful where position data is unavailable, but holdings data is available here and is far more precise.

How do you stop the risk model from becoming a constraint that kills alpha?

Limits target unintended exposures, not intended ones. The cost of each hedge in expected alpha is measured, and limits are designed together with PMs. The framework should help PMs keep the alpha they want and shed the risk they do not.

How do you measure crowding?

At three levels: a short-side squeeze factor in the core model built mainly from prime broker borrow data, return-based crowding indicators for each factor in the monitoring layer, and internal overlap across our own teams for liquidity stress. The last one is often the most important, because no single team can see it.

What would a purely cross-sectional model miss for intraday strategies?

Market-wide shocks. A book of reversal strategies is effectively short volatility and short liquidity, and multi-market books carry currency, rates and US-session exposure, while every cross-sectional exposure can look neutral. I add time-series shock factors, use index futures as the intraday market factor, and correct betas for asynchronous trading.

How often should exposures update?

Positions in real time. Price-based exposures (reversal, momentum, volatility, liquidity) intraday or daily; slow characteristics daily. The factor covariance is re-estimated daily, with short half-lives.

Why build in-house rather than buy a vendor model?

The core factors for short holding periods come from price, volume and position data the firm already has, so most of the model can be built without a licence, and it fits our horizons better than a generic model. The cost is build effort and the lack of an external audit, which I would offset with a PCA challenger, a simple baseline and stronger internal validation.

How do you decide whether to add or remove a factor?

Marginal explanatory power, reduction in residual correlation across teams, stability of factor returns, and whether anyone can act on it. Changes go through a documented governance process, because changing definitions breaks the comparability of history.

Appendix B: References

Grouped by the part of the framework they support. The framework itself is a synthesis; no single paper proposes it as a whole.

Linear factor risk models and estimation

Menchero, J., Orr, D. J. and Wang, J. (2011). The Barra US Equity Model (USE4): Methodology Notes. MSCI.
Cross-sectional factor model design, eigenfactor risk adjustment and volatility regime adjustment (Sections 3.2 and 3.4).

Grinold, R. C. and Kahn, R. N. (1999). Active Portfolio Management, 2nd edition. McGraw-Hill.
Standard reference for factor risk models, risk decomposition and performance attribution.

Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. Journal of Multivariate Analysis, 88(2), 365-411.
Covariance shrinkage (Section 3.4).

French, K. R. Data Library. Tuck School of Business, Dartmouth College. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Public factor return series, used as an external sanity check on in-house slow-factor definitions (Section 3.6).

Sharpe, W. F. (1992). Asset allocation: Management style and performance measurement. Journal of Portfolio Management, 18(2), 7-19.

Returns-based style analysis, used here as a cross-check to holdings-based attribution.

Statistical and hybrid factor models

Connor, G. and Korajczyk, R. A. (1986). Performance measurement with the arbitrage pricing theory: A new framework for analysis. Journal of Financial Economics, 15(3), 373-394.

Statistical (principal component) factor models.

Fan, J., Liao, Y. and Mincheva, M. (2013). Large covariance estimation by thresholding principal orthogonal complements. Journal of the Royal Statistical Society, Series B, 75(4), 603-680.

Factor structure plus sparse residual covariance; support for statistical residual factors.

Kelly, B., Pruitt, S. and Su, Y. (2019). Characteristics are covariances: A unified model of risk and return. Journal of Financial Economics, 134(3), 501-524.

IPCA: latent factors with characteristic-dependent loadings, bridging named and statistical factors.

Lettau, M. and Pelger, M. (2020). Factors that fit the time series and cross-section of stock returns. Review of Financial Studies, 33(5), 2274-2325.

Risk-premium PCA; statistical factors estimated with pricing information.

Machine learning and stock returns

Gu, S., Kelly, B. and Xiu, D. (2020). Empirical asset pricing via machine learning. Review of Financial Studies, 33(5), 2223-2273.

The most important predictors in ML return models are short-term reversal, momentum, liquidity and volatility, which supports the core factor set (Section 2.2).

Avramov, D., Cheng, S. and Metzker, L. (2023). Machine learning versus economic restrictions: Evidence from stock return predictability. Management Science, 69(5), 2587-2619.

ML signals concentrate in hard-to-trade stocks; relevant to liquidity and cost attribution.

Gu, S., Kelly, B. and Xiu, D. (2021). Autoencoder asset pricing models. Journal of Econometrics, 222(1), 429-450.

Nonlinear (deep) factor model: the main alternative to a linear core.

Chen, L., Pelger, M. and Zhu, J. (2024). Deep learning in asset pricing. Management Science, 70(2), 714-750.

Deep-learning factor model: the main alternative to a linear core.

Explainability and its limits

Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems 30.

Introduces SHAP, used in the surrogate diagnostic (Section 3.2).

Ribeiro, M. T., Singh, S. and Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. Proceedings of ACM KDD.

Introduces LIME, a perturbation-based local explainer.

Man Group (Man Numeric). Shining Light into the Machine Learning Black Box. Man Group Insights. https://www.man.com/insights/shining-light-machine-learning

Practitioner precedent: SHAP with tree models to attribute both portfolio weights and returns. Goes further than this framework by using SHAP as the attribution itself.

Chen, D. (2025). Explaining risks: Axiomatic risk attributions for financial models. Quantitative Finance, 25(6), 1007-1014.

Extends Shapley values from explaining predictions to allocating risk.

Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, 1(5), 206-215.

Argument for an interpretable core rather than post-hoc explanation of a complex model.

Slack, D., Hilgard, S., Jia, E., Singh, S. and Lakkaraju, H. (2020). Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, 180-186.

Shows that perturbation-based explanations can be manipulated by a model that detects synthetic inputs.

Dombrowski, A.-K., Alber, M., Anders, C. J., Ackermann, M., Müller, K.-R. and Kessel, P. (2019). Explanations can be manipulated and geometry is to blame. Advances in Neural Information Processing Systems 32.

Gradient-based explanations can be changed by imperceptible input changes without changing predictions.

Kumar, I. E., Venkatasubramanian, S., Scheidegger, C. and Friedler, S. (2020). Problems with Shapley-value-based explanations as feature importance measures. Proceedings of ICML, PMLR 119.

Conceptual problems: off-distribution evaluation, credit to unused correlated features, and lack of actionability.

Sundararajan, M. and Najmi, A. (2020). The many Shapley values for model explanation. Proceedings of ICML, PMLR 119.

Different Shapley formulations give different attributions for the same model.

Merrick, L. and Taly, A. (2020). The explanation game: Explaining machine learning models using Shapley values. CD-MAKE 2020, Lecture Notes in Computer Science 12279.

Shows how game formulation changes attributions, and proposes confidence intervals for them.

Intraday and microstructure factors

Nagel, S. (2012). Evaporating liquidity. Review of Financial Studies, 25(7).

Short-term reversal returns as compensation for liquidity provision, linked to implied volatility (volatility and liquidity shock factors).

Pástor, Ľ. and Stambaugh, R. F. (2003). Liquidity risk and expected stock returns. Journal of Political Economy, 111(3).

Market-wide liquidity as a time-series risk factor.

Epps, T. W. (1979). Comovements in stock prices in the very short run. Journal of the American Statistical Association, 74(366).

High-frequency correlations biased toward zero (Section 3.4).

Hayashi, T. and Yoshida, N. (2005). On covariance estimation of non-synchronously observed diffusion processes. Bernoulli, 11(2).

Covariance estimator for asynchronous intraday data.

Ben-David, I., Franzoni, F. and Moussawi, R. (2018). Do ETFs increase volatility? Journal of Finance, 73(6).

ETF ownership and stock volatility (ETF and index-basket co-movement).

Chordia, T. and Subrahmanyam, A. (2004). Order imbalance and individual stock returns: Theory and evidence. Journal of Financial Economics, 72(3).

Order imbalance and short-horizon returns.

Cont, R., Kukanov, A. and Stoikov, S. (2014). The price impact of order book events. Journal of Financial Econometrics, 12(1).

Order flow imbalance explains short-interval price changes.

Boehmer, E., Jones, C. M., Zhang, X. and Zhang, X. (2021). Tracking retail investor activity. Journal of Finance, 76(5).

Identifying US retail trades from sub-penny price improvement.

Heston, S. L., Korajczyk, R. A. and Sadka, R. (2010). Intraday patterns in the cross-section of stock returns. Journal of Finance, 65(4).

Intraday return periodicity; treated as alpha, not risk.

Gao, L., Han, Y., Li, S. Z. and Zhou, G. (2018). Market intraday momentum. Journal of Financial Economics, 129(2).

First half-hour return predicts last half-hour; treated as alpha, not risk.

Diversification, crowding and governance

López de Prado, M. (2016). Building diversified portfolios that outperform out of sample. Journal of Portfolio Management, 42(4), 59-69.

Hierarchical clustering of return correlations, relevant to detecting overlap between teams.

Lou, D. and Polk, C. (2022). Comomentum: Inferring arbitrage activity from return correlations. Review of Financial Studies.

Return-correlation measure of strategy crowding, adapted here for factor-level crowding indicators (Section 2.2).

Hanson, S. G. and Sunderam, A. (2014). The growth and limits of arbitrage: Evidence from short interest. Review of Financial Studies, 27(4).

Short interest in a factor's short leg as a signal of arbitrage capital in that factor.

Khandani, A. E. and Lo, A. W. (2007). What happened to the quants in August 2007? Journal of Investment Management, 5(4), 5-54.

Crowding and deleveraging in quant equity strategies (Sections 2.2 and 3.6).

Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency (2011). Supervisory Guidance on Model Risk Management (SR 11-7).

Effective challenge and independent validation; supports keeping the risk model independent of the alpha models.